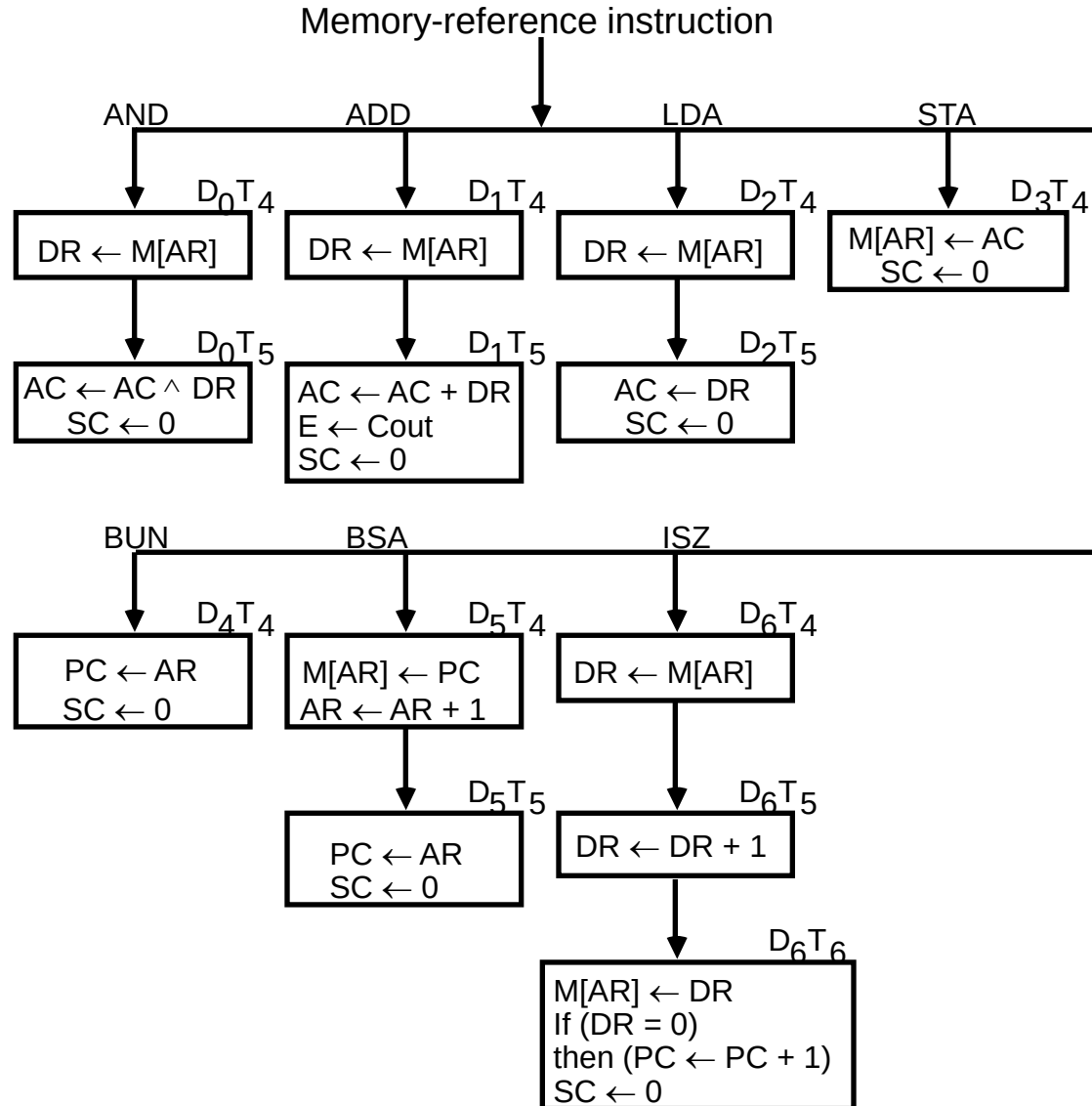


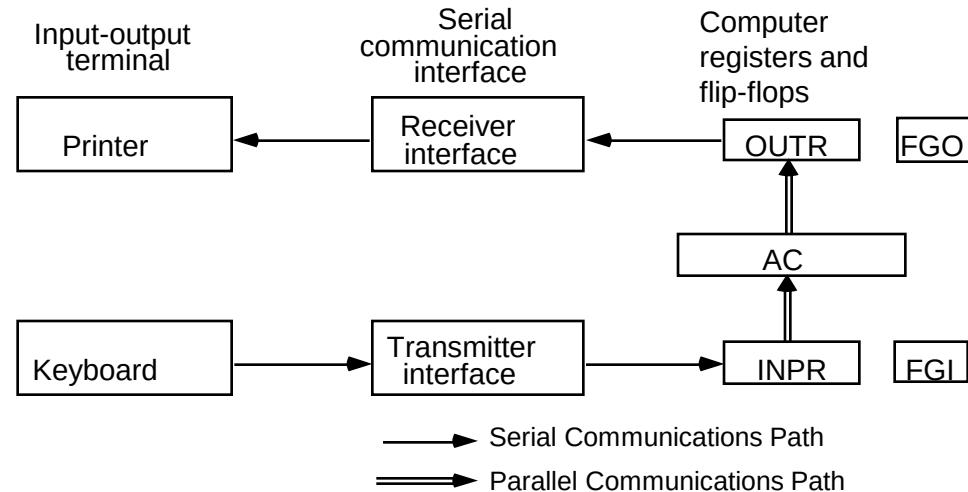
FLOWCHART FOR MEMORY REFERENCE INSTRUCTIONS



INPUT-OUTPUT AND INTERRUPT

A Terminal with a keyboard and a Printer

• Input-Output Configuration



INPR Input register - 8 bits
OUTR Output register - 8 bits
FGI Input flag - 1 bit
FGO Output flag - 1 bit
IEN Interrupt enable - 1 bit

- The terminal sends and receives serial information
- The serial info. from the keyboard is shifted into INPR
- The serial info. for the printer is stored in the OUTR
- INPR and OUTR communicate with the terminal serially and with the AC in parallel.
- The flags are needed to *synchronize* the timing difference between I/O device and the computer



PROGRAM CONTROLLED DATA TRANSFER

-- CPU --

/* Input */ /* Initially FGI = 0 */

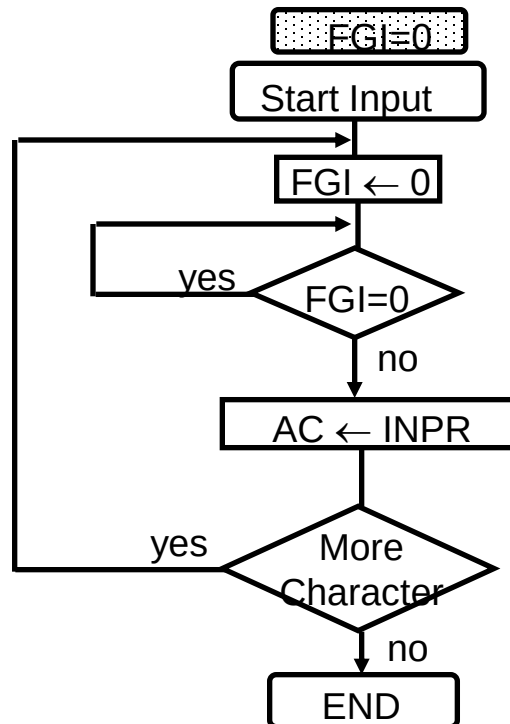
loop: If FGI = 0 goto loop

AC ← INPR, FGI ← 0

/* Output */ /* Initially FGO = 1 */

loop: If FGO = 0 goto loop

OUTR ← AC, FGO ← 0



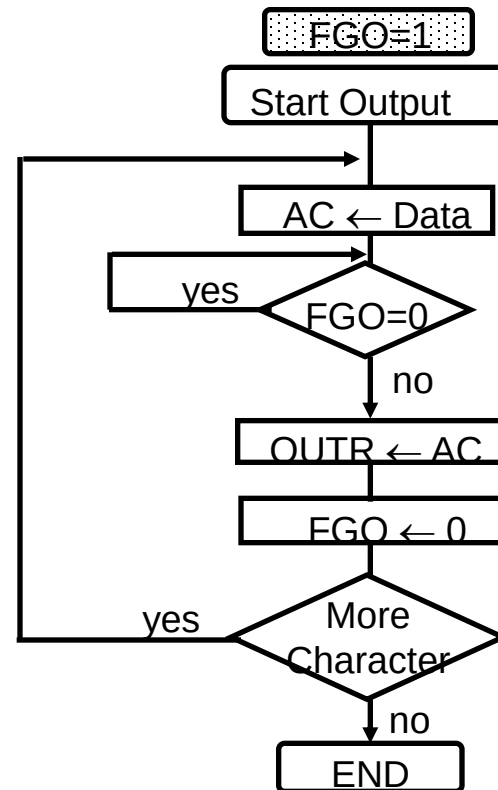
-- I/O Device --

loop: If FGI = 1 goto loop

INPR ← new data, FGI ← 1

loop: If FGO = 1 goto loop

consume OUTR, FGO ← 1



INPUT-OUTPUT INSTRUCTIONS

$$D_7IT_3 = p$$

$$IR(i) = B_i, i = 6, \dots, 11$$

INP	p:	$SC \leftarrow 0$	Clear SC
OUT	pB_{11} :	$AC(0-7) \leftarrow INPR, FGI \leftarrow 0$	Input char. to AC
SKI	pB_{10} :	$OUTR \leftarrow AC(0-7), FGO \leftarrow 0$	Output char. from AC
SKO	pB_9 :	if($FGI = 1$) then ($PC \leftarrow PC + 1$)	Skip on input flag
ION	pB_8 :	if($FGO = 1$) then ($PC \leftarrow PC + 1$)	Skip on output flag
IOF	pB_7 :	$IEN \leftarrow 1$	Interrupt enable on
	pB_6 :	$IEN \leftarrow 0$	Interrupt enable off



PROGRAM-CONTROLLED INPUT/OUTPUT

- Program-controlled I/O
 - Continuous CPU involvement
I/O takes valuable CPU time
 - CPU slowed down to I/O speed
 - Simple
 - Least hardware

Input

```
LOOP,   SKI   DEV
        BUN   LOOP
        INP   DEV
```

Output

```
LOOP,   LDA   DATA
LOP,    SKO   DEV
        BUN   LOP
        OUT   DEV
```



INTERRUPT INITIATED INPUT/OUTPUT

- Open communication only when some data has to be passed --> *interrupt*.
- The I/O interface, instead of the CPU, monitors the I/O device.
- When the interface finds that the I/O device is ready for data transfer, it generates an interrupt request to the CPU
- Upon detecting an interrupt, the CPU stops momentarily the task it is doing, branches to the service routine to process the data transfer, and then returns to the task it was performing.

* IEN (Interrupt-enable flip-flop)

- can be set and cleared by instructions
- when cleared, the computer cannot be interrupted

